what the heck, let's just say it—shooting him? Don't you have free will? Here is where, some suspect, quantum mechanics may enter the story.

Free Will, Many Worlds, and Time Travel

Free will is a tricky issue, even absent the complicating factor of time travel. The laws of classical physics are deterministic. As we saw earlier, if you were to know precisely how things are now (the position and velocity of every particle in the universe), the laws of classical physics would tell you exactly how things were or would be at any other moment you specified. The equations are indifferent to the supposed freedom of human will. Some have taken this to mean that in a classical universe, free will would be an illusion. You are made of a collection of particles, so if the laws of classical physics could determine everything about your particles at any moment—where they'd be, how they'd be moving and so on—your willful ability to determine your own actions would appear fully compromised. This reasoning convinces me, but those who believe we are more than the sum of our particles may disagree.

Anyway, the relevance of these observations is limited, since ours is a quantum, not a classical, universe. In quantum physics, real-world physics, there are resemblances to this classical perspective; there are also potentially pivotal differences. As you read in Chapter 7, if you know the quantum wavefunction right now for every particle in the universe, Schrödinger's equation tells you how the wavefunction was or will be at any other moment you specify. This component of quantum physics is fully deterministic, just as in classical physics. However, the act of observation complicates the quantum mechanical story and, as we've seen, heated debate over the quantum measurement problem still rages. If physicists one day conclude that Schrödinger's equation is all there is to quantum mechanics, then quantum physics, in its entirety, would be every bit as deterministic as classical physics. As with classical determinism, some would say this means free will is an illusion; others would not. But if we're currently missing part of the quantum story—if the passage from probabilities to definite outcomes requires something beyond the standard quantum framework—it's at least possible that free will might find a concrete realization within physical law. We might one day find, as some physicists have speculated, that the act of conscious